

Sealed Experiment Record

Sealed 2026-10-10 · Signed by the K-Veritas server
Session 1cb621b5 · Data hash 1ca0...d5e2

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-10 19:23 and 19:29 UTC. Reported results: lambada_openai_perplexity 8.13741, lambada_openai_acc 0.556375, hellaswag_acc 0.372037, hellaswag_acc_norm 0.464848. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	1cb621b5-c52c-46ab-b837-9da028960e3b
Initialized	2026-10-10T19:23:57Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	a51cc1e62f20c5fb305dfcc132e4ad52fe231867bc2493ad3b06d81584471322
Disclosure	open
Source code, run 1	ab9c9ad34d63601e50b0dc711394130ccd2b2eb66b805de5f5391e00ebcd2305

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	5 minutes 15 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
lambada_openai_sample_len	5153		explicit
lambada_openai_perplexity	8.13741		explicit
lambada_openai_acc	0.556375		explicit

Metric	Final value	Step	Source
hellaswag_sample_len	10042		explicit
hellaswag_acc	0.372037		explicit
hellaswag_acc_norm	0.464848		explicit
piqa_sample_len	1838		explicit
piqa_acc	0.695321		explicit
piqa_acc_norm	0.683351		explicit
arc_easy_sample_len	2376		explicit
arc_easy_acc	0.549242		explicit
arc_easy_acc_norm	0.481902		explicit
arc_challenge_sample_len	1172		explicit
arc_challenge_acc	0.25		explicit
arc_challenge_acc_norm	0.277304		explicit
winogrande_sample_len	1267		explicit
winogrande_acc	0.555643		explicit

Inline claim	Value	Phase	Output line
lambada_openai_perplexity	8.13741		17
lambada_openai_acc	0.556375		19
hellaswag_acc	0.372037		22
hellaswag_acc_norm	0.464848		24
piqa_acc	0.695321		27
piqa_acc_norm	0.683351		29
arc_easy_acc	0.549242		32
arc_easy_acc_norm	0.481902		34
arc_challenge_acc	0.25		37
arc_challenge_acc_norm	0.277304		39
winogrande_acc	0.555643		42

4 Execution evidence

Coherence

Coherent (0.36): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	3136	374.2 s	37.67 GB	97%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 84 tracked files, 3 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	19:23:57	95039123606bbfd1	84
2	run_end	19:29:13	95039123606bbfd1	0

1 snapshot with no file change omitted here; full timeline in Appendix B.

Withheld by .kveritasignore	Size	Hash
csrc/selective_scan/selective_scan.cpp	<1MB	51c778938926eda04794413a

Withheld by .kveritasignore	Size	Hash
csrc/selective_scan/selective_scan.h	<1MB	e5042ee475aa0146914fa465
csrc/selective_scan/selective_scan_common.h	<1MB	50f7b4ddcc6cc9e4c692a56
csrc/selective_scan/static_switch.h	<1MB	a339beb3d7bc98522a181475

Activity: 85 files read, 0 files written, 33 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-10T19:29:14.657074+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	1ca0f6d826b86b3262aedbf1866626edd8c23b0e04c825c7feb92fd61ead5e2
Code bundle hash	7b483483fd2b5bf57293081c8076c98b1aa147020fc7bae55c1a1be7f43cfc6

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
lambada_openai_sample_len	5153		15	explicit
lambada_openai_perplexity	8.13741		16	explicit
lambada_openai_acc	0.556375		18	explicit
hellaswag_sample_len	10042		20	explicit
hellaswag_acc	0.372037		21	explicit
hellaswag_acc_norm	0.464848		23	explicit
piqa_sample_len	1838		25	explicit
piqa_acc	0.695321		26	explicit
piqa_acc_norm	0.683351		28	explicit
arc_easy_sample_len	2376		30	explicit
arc_easy_acc	0.549242		31	explicit
arc_easy_acc_norm	0.481902		33	explicit
arc_challenge_sample_len	1172		35	explicit
arc_challenge_acc	0.25		36	explicit
arc_challenge_acc_norm	0.277304		38	explicit
winogrande_sample_len	1267		40	explicit
winogrande_acc	0.555643		41	explicit

Phase	Line	Time	CPU time	Memory	GPU
evaluate	1	19:23:58	1.4 s	0.13 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	19:23:57	95039123606bbfd1	84
1	phase evaluate	19:23:58	95039123606bbfd1	0
2	run_end	19:29:13	95039123606bbfd1	0

Event	Path or command	Hash
read	kv_run.sh	
read	.kveritasignore	
read	.github/scripts/build.sh	
read	.github/scripts/check_for_ngc_images.sh	
read	.github/scripts/check_for_releases.sh	
read	.github/scripts/test.sh	
read	.github/workflows/_build.yml	
read	.github/workflows/_build_in_container.yml	
read	.github/workflows/build.yml	
read	.github/workflows/build_in_container.yml	
read	.github/workflows/build_in_container_scheduled.yml	
read	.github/workflows/publish.yml	
read	benchmarks/benchmark_generation_mamba_simple.py	

Event	Path or command	Hash
read	csrc/selective_scan/selective_scan.cpp	
read	csrc/selective_scan/selective_scan.h	
read	csrc/selective_scan/selective_scan_common.h	
read	csrc/selective_scan/static_switch.h	
read	evals/lm_harness_eval.py	
read	mamba_ssm/__init__.py	
read	mamba_ssm/distributed/__init__.py	
read	mamba_ssm/distributed/distributed_utils.py	
read	mamba_ssm/distributed/tensor_parallel.py	
read	mamba_ssm/models/__init__.py	
read	mamba_ssm/models/config_mamba.py	
read	mamba_ssm/models/mixer_seq_simple.py	
read	mamba_ssm/modules/__init__.py	
read	mamba_ssm/modules/block.py	
read	mamba_ssm/modules/mamba2.py	
read	mamba_ssm/modules/mamba2_simple.py	
read	mamba_ssm/modules/mamba3.py	
read	mamba_ssm/modules/mamba_simple.py	
read	mamba_ssm/modules/mha.py	
read	mamba_ssm/modules/mlp.py	
read	mamba_ssm/modules/ssd_minimal.py	
read	mamba_ssm/ops/__init__.py	
read	mamba_ssm/ops/cute/__init__.py	
read	mamba_ssm/ops/cute/mamba3/__init__.py	
read	mamba_ssm/ops/cute/mamba3/mamba3_step_fn.py	
read	mamba_ssm/ops/selective_scan_interface.py	
read	mamba_ssm/ops/tilelang/__init__.py	
read	mamba_ssm/ops/tilelang/mamba3/__init__.py	
read	mamba_ssm/ops/tilelang/mamba3/mamba3_mimo.py	
read	mamba_ssm/ops/tilelang/mamba3/mamba3_mimo_bwd.py	
read	mamba_ssm/ops/tilelang/mamba3/mamba3_mimo_bwd_varlen.py	
read	mamba_ssm/ops/tilelang/mamba3/mamba3_mimo_fwd.py	
read	mamba_ssm/ops/tilelang/mamba3/mamba3_mimo_fwd_varlen.py	
read	mamba_ssm/ops/triton/__init__.py	
read	mamba_ssm/ops/triton/k_activations.py	
read	mamba_ssm/ops/triton/layer_norm.py	
read	mamba_ssm/ops/triton/layernorm_gated.py	
read	mamba_ssm/ops/triton/mamba3/__init__.py	
read	mamba_ssm/ops/triton/mamba3/angle_dt.py	
read	mamba_ssm/ops/triton/mamba3/grouped_head_reduction.py	
read	mamba_ssm/ops/triton/mamba3/mamba3_mimo_rotary_step.py	
read	mamba_ssm/ops/triton/mamba3/mamba3_mimo_utils.py	
read	mamba_ssm/ops/triton/mamba3/mamba3_asiso_bwd.py	
read	mamba_ssm/ops/triton/mamba3/mamba3_asiso_combined.py	
read	mamba_ssm/ops/triton/mamba3/mamba3_asiso_fwd.py	
read	mamba_ssm/ops/triton/mamba3/mamba3_asiso_step.py	
read	mamba_ssm/ops/triton/mamba3/utils.py	
read	mamba_ssm/ops/triton/selective_state_update.py	
read	mamba_ssm/ops/triton/softplus.py	
read	mamba_ssm/ops/triton/ssd_bmm.py	

Event	Path or command	Hash
read	mamba_ssm/ops/triton/ssd_chunk_scan.py	
read	mamba_ssm/ops/triton/ssd_chunk_state.py	
read	mamba_ssm/ops/triton/ssd_combined.py	
read	mamba_ssm/ops/triton/ssd_state_passing.py	
read	mamba_ssm/utils/__init__.py	
read	mamba_ssm/utils/determinism.py	
read	mamba_ssm/utils/generation.py	
read	mamba_ssm/utils/hf.py	
read	mamba_ssm/utils/torch.py	
read	pyproject.toml	
read	setup.py	
read	tests/benchmark_determinism_kernels.py	
read	tests/modules/test_mamba3_varlen.py	
read	tests/ops/cute/test_mamba3_mimo_step.py	
read	tests/ops/test_selective_scan.py	
read	tests/ops/tilelang/test_mamba3_mimo.py	
read	tests/ops/triton/test_layernorm_gated.py	
read	tests/ops/triton/test_mamba3_asiso.py	
read	tests/ops/triton/test_selective_state_update.py	
read	tests/ops/triton/test_ssdl.py	
read	tests/test_determinism.py	
read	tests/test_generation.py	
process	grep -E ^\	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	pid 2740302	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/lib/python3.12/site-packages/t...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/lib/python3.12/site-packages/t...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/var/tmp/kv-repro/032-state-spaces-mamba/env/bin/python /var/tmp/kv-repro/...	
process	/bin/sh -c nvidia-smi -L	
process	pid 2748521	
process	/bin/sh -c nvidia-smi	
process	nvidia-smi	
process	pid 2748545	
process	pid 2748544	

Event	Path or command	Hash
process	pid 2748564	
process	/bin/sh -c clang --version	
process	/bin/sh -c lscpu	
process	lscpu	
process	python - /var/tmp/kv-repro/032-state-spaces-mamba/data/results-1791660237	

C Run history

Invocation 1, OK, 5 minutes 15 seconds, 42 output lines, started 2026-10-10T19:23:57

metric_hash a0679a393c4e57223e51d1322c21a9a41ffc9be88662d893238156f61e1c008f
run_digest 891e7317fe5c34b25eca388845f5d2c32201db322883cc71aaf9d40f6c8ed552
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 1ca0f6d826b86b3262aedbf1866626edd8c23b0e04c825c7febf92fd61ead5e2
signed_message_hash c580d8e56c4b93e60a264bf28e94960b28eab0971e2bffe47f4b4bf86ec51b87
nonce 73ea081e187eaf5996a1d9897ac1a1b2
signed_at 2026-10-10T19:29:14.657074+00:00
sealed_at 2026-10-10T19:29:14.715176357Z
checkout_bundle 7b483483fd2b5bf57293081c8076c98b1aa147020fc7bae55c1a1be7f43cfc6

Signature (base64)

fIFCClANv+rYCC6hv5ggWUMAPCD9qMLzzL4bac+FHpa1fAczJM9EL7JpvgHuNcd2cFkpTE4fIBAdHgFDRD8vgnDv
frwddLfmyOB0qvYpXU/HjFpCrOhIO8N/AtL9vmAYtPpM9BS0Euibpg/62L/axVynmO46nOp8LWy0L/WTH8ZZbN7P
HcxM0kK6oJ3f/QfHQ7k0gNMF8wCZz4gQzZZJTMFLhv0h0fhRCuBG/SBdv3k3Vn4Conz+ta534mLbphXEFrLH4hF
ghN+Mmd1eQxPXuqK/ZchvY0uUr4Tm3JDjahzyZu5KETJVz+MQLsFTlWkoIOqJgbG8K6JkMBzWSg8FLTyww418TKw
uvFfG5w0m1RHU8w8692bldYUcC0pGLFzqwDfj0LVZCzjbo77IU/J5/bLc1bzh/j+NN0cXdVxwkbUJ1xt7hR5f80S
BhLLKyhZm7+ZYwbsN016TyCxJof4wtP3400TEwWacUu4s30CawUkb+gRNZ688W7ck7bbvcBAFGuHY6qA/OhXS1i
MXNEuaV3TmCjSaHU7lr1QnU7ILG8X5VPbM1fYE4nxoioBLWSrRnMx8xGFgjWPUQR1A20hK0UnSLFyyER+Owglobo
04XE7LtyzaHEOUSeC4pwCg7UADeLvsNg/GRJbT8mxdKymbsEAJZCL4UmJrwrYL6B1cU=

Public key

-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEA1QL3YyMgeShx8IhtC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlP1cJayJxD5fcPrOPPCqp/LoUFz5CUZzRi
6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNpPxYa5X
7qu95EfEisX/GfoS9ToxtqPBWcCvSPBX8efec4TC58m96Fsp4mC8KicEadbIbHN
av0JjdTJ//CMOWZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdq+D
AhfTRuJNgWEkjwD5lb0Vhi4qlkweSOEMTKT3ye9I15xEaJn4bku4eKqes7VSZt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHNcy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtf04R5UGVV92qd2Mi
1GN7lE3mbQQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PwjncNku5cdw3gHLCCAwEAAQ==
-----END PUBLIC KEY-----

Run 1 outputs

stdout 419bf08a84ed9922d022ead5741138df87fd2935a42ba4c565483e99cc0ad1f
stderr e3b0c44298fc1c149afbfc4c8996fb92427ae41e4649b934ca495991b7852b855
metrics a0679a393c4e57223e51d1322c21a9a41ffc9be88662d893238156f61e1c008f
digest 891e7317fe5c34b25eca388845f5d2c32201db322883cc71aaf9d40f6c8ed552
kv_run.sh (unchanged)
pre 9bb15dcd0a238b12aefec13f2cc1f6c844067e871be863e449a2eb1898225e44
post 9bb15dcd0a238b12aefec13f2cc1f6c844067e871be863e449a2eb1898225e44

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
```